



**BITS Pilani**  
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# Automotive Communication Systems Lab

## Experiment 1 – DBC file creation using TS Master

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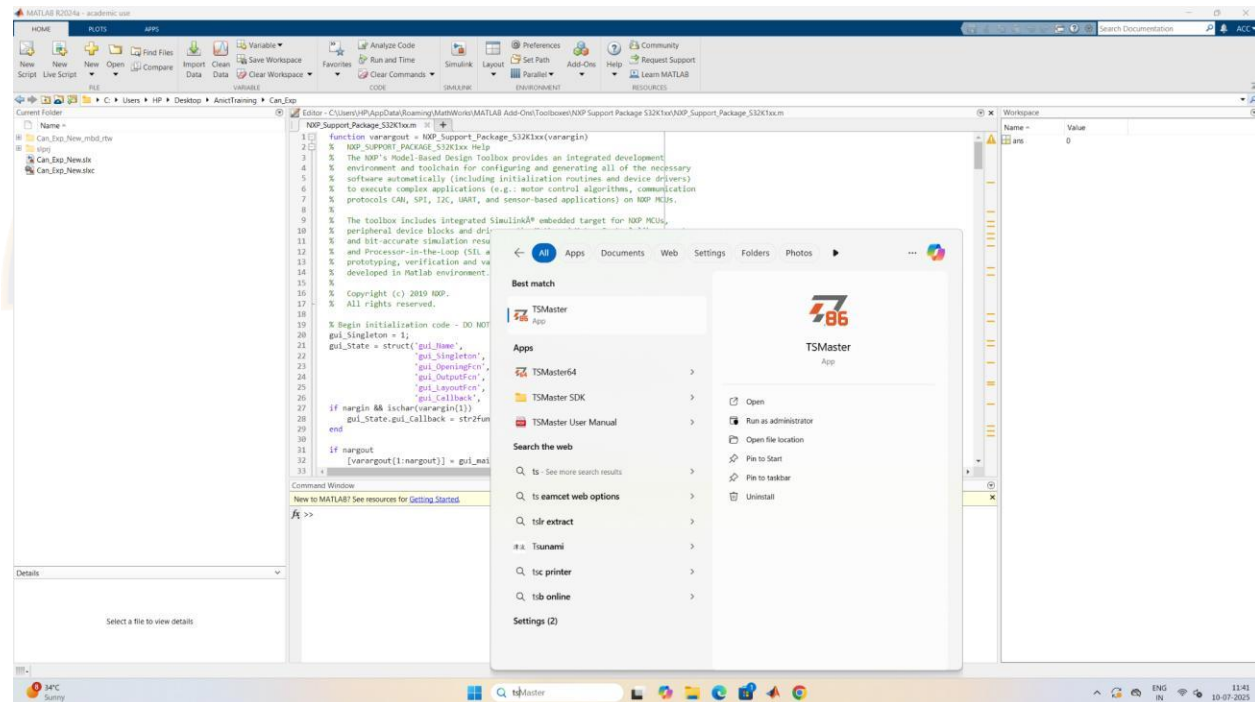


# Exercise-1

# TS MASTER

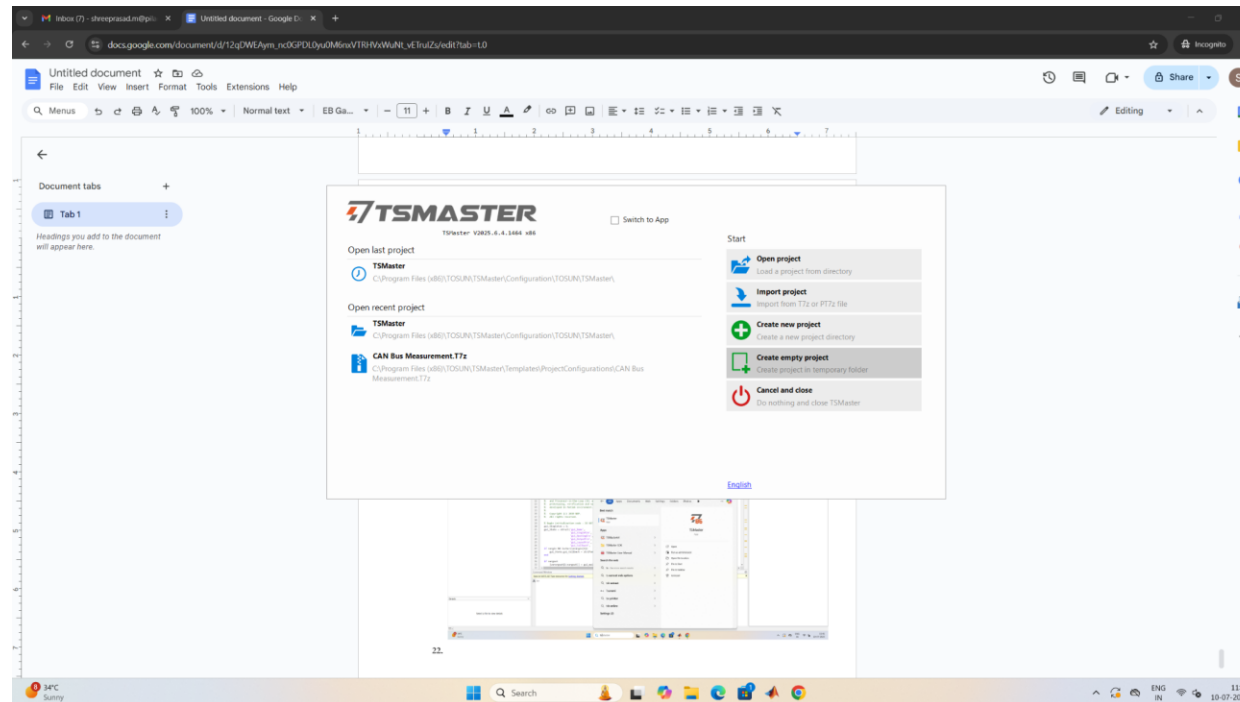
❑ Download TS Master Application

❑ Open it



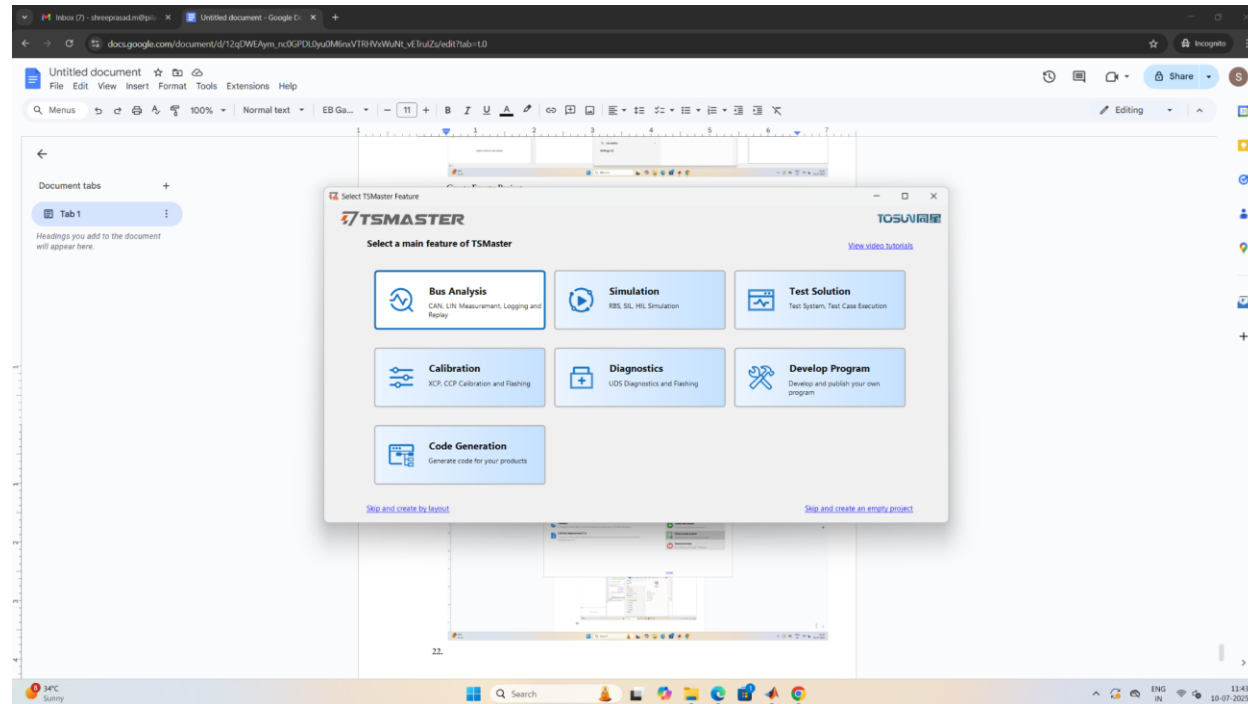
# TS MASTER

## ❑ Create empty project



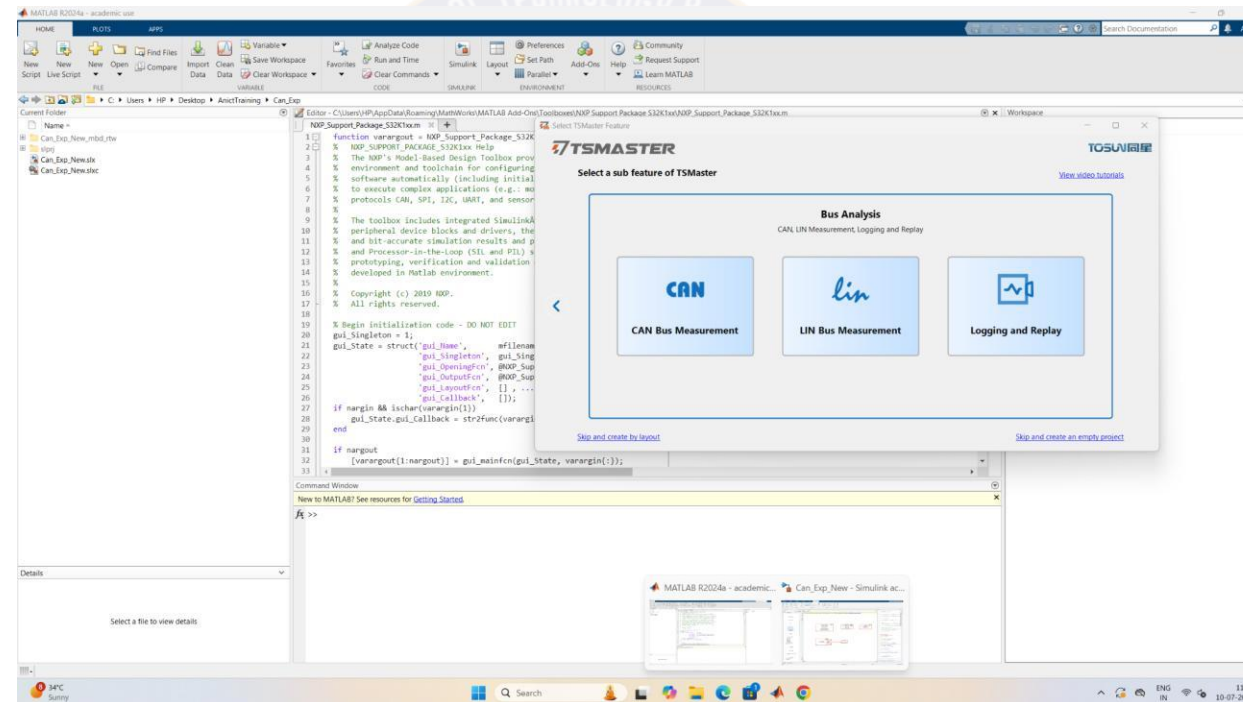
# TS MASTER

## ❑ Select Bus Analysis



# TS MASTER

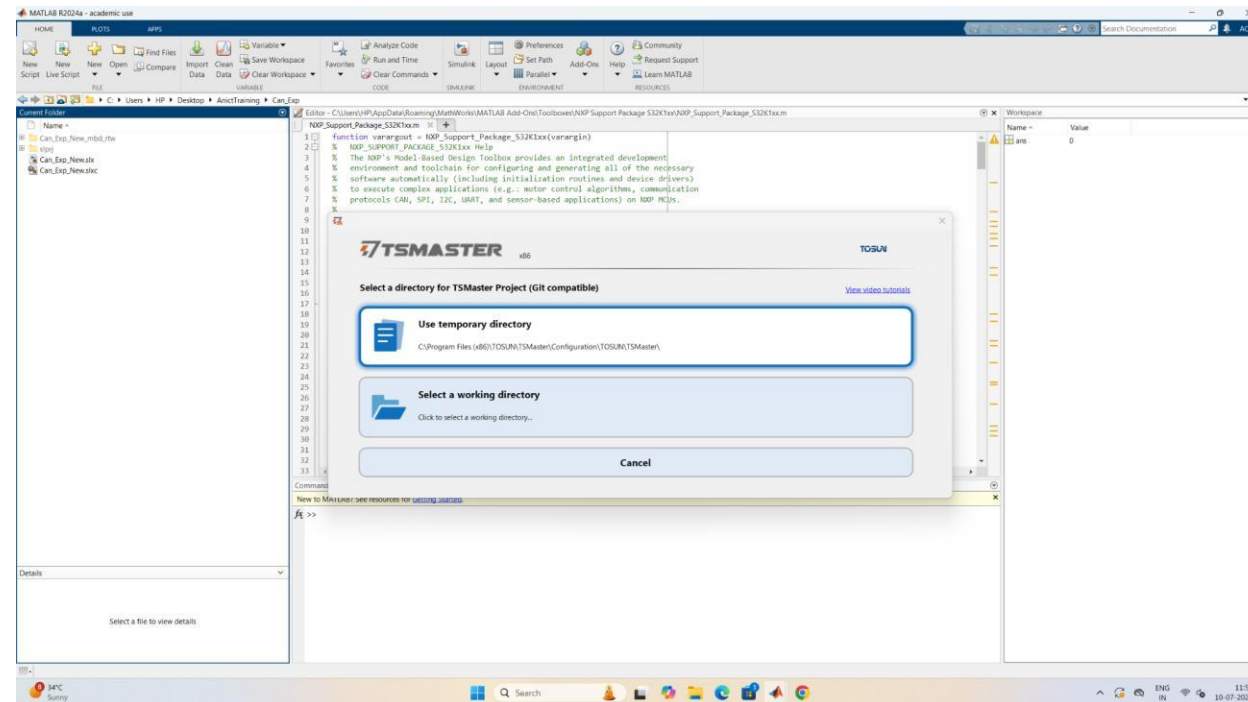
## ❑ Select CAN bus measurement





# TS MASTER

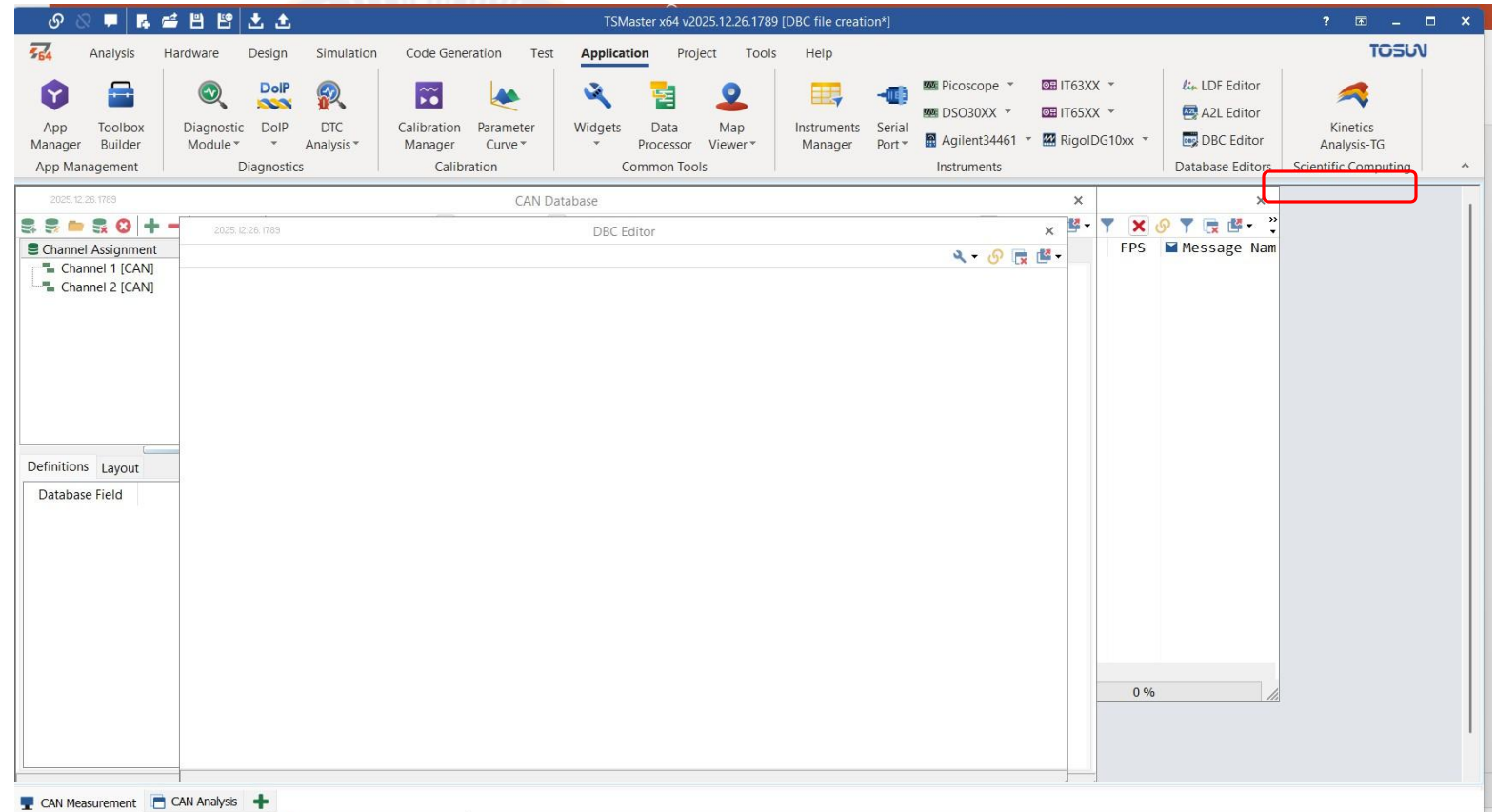
## ❑ Select temporary directory



# TS MASTER

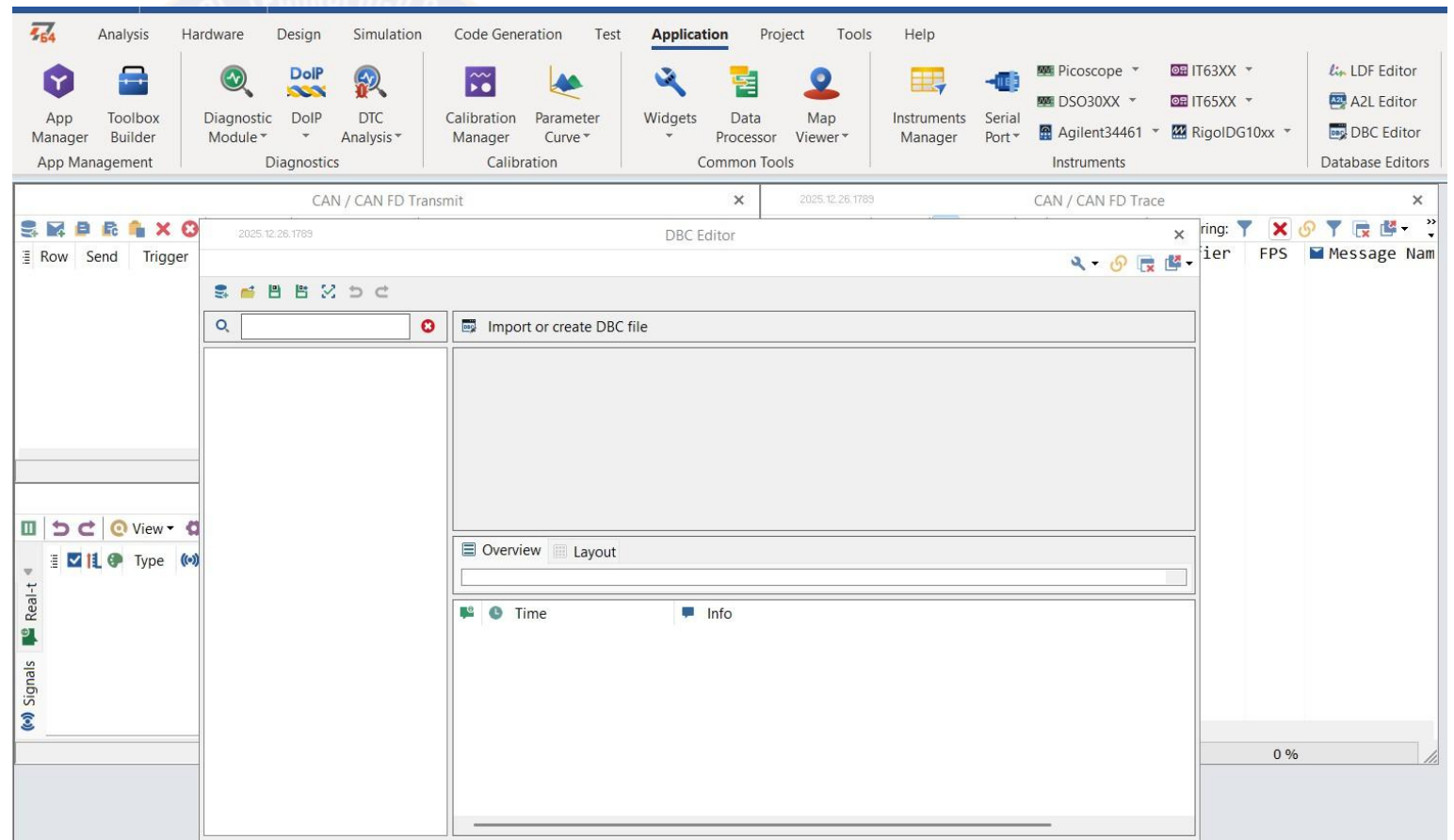
❑ Go to Application

❑ Click DBC Editor





# TS MASTER

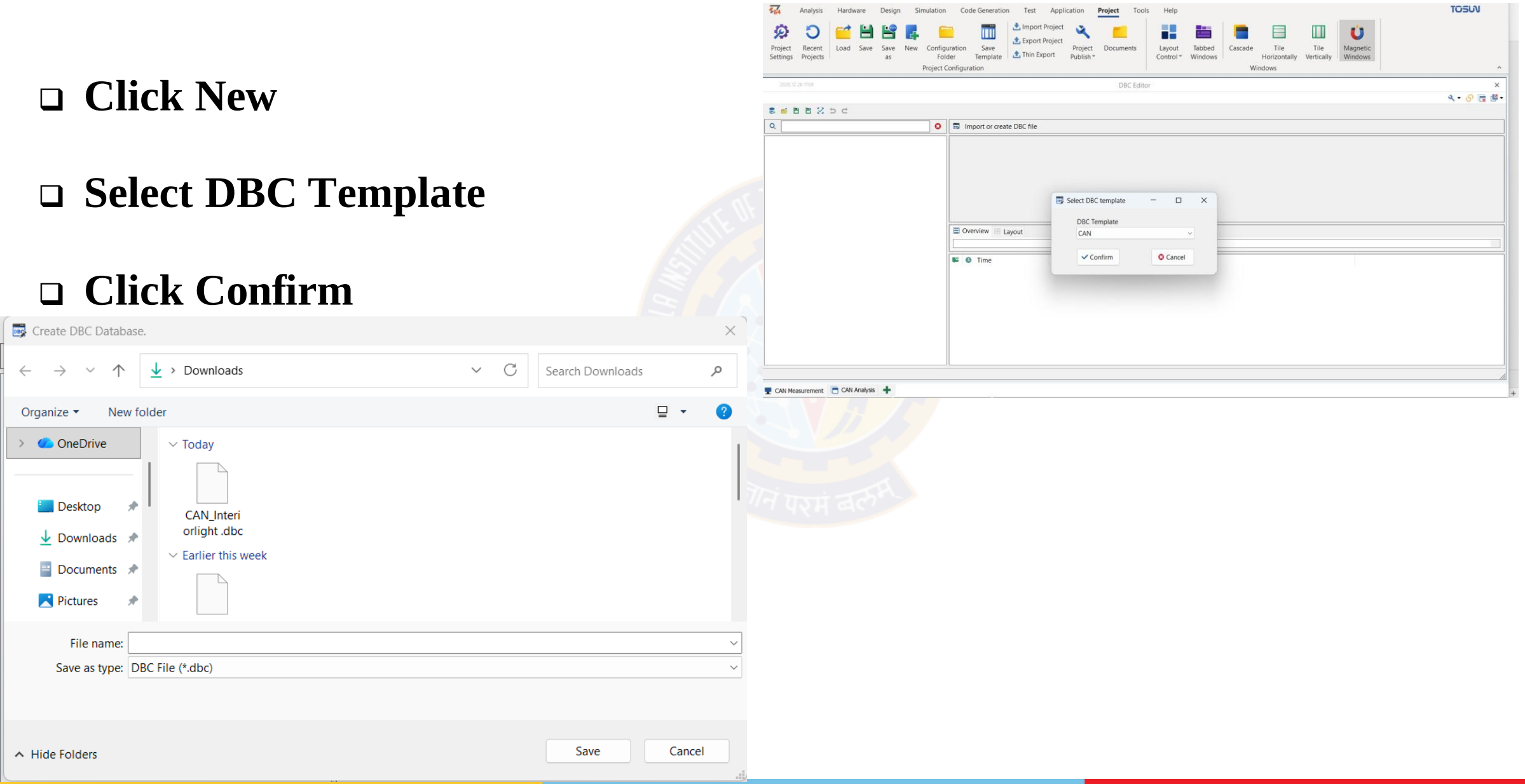


# TS MASTER

❑ Click New

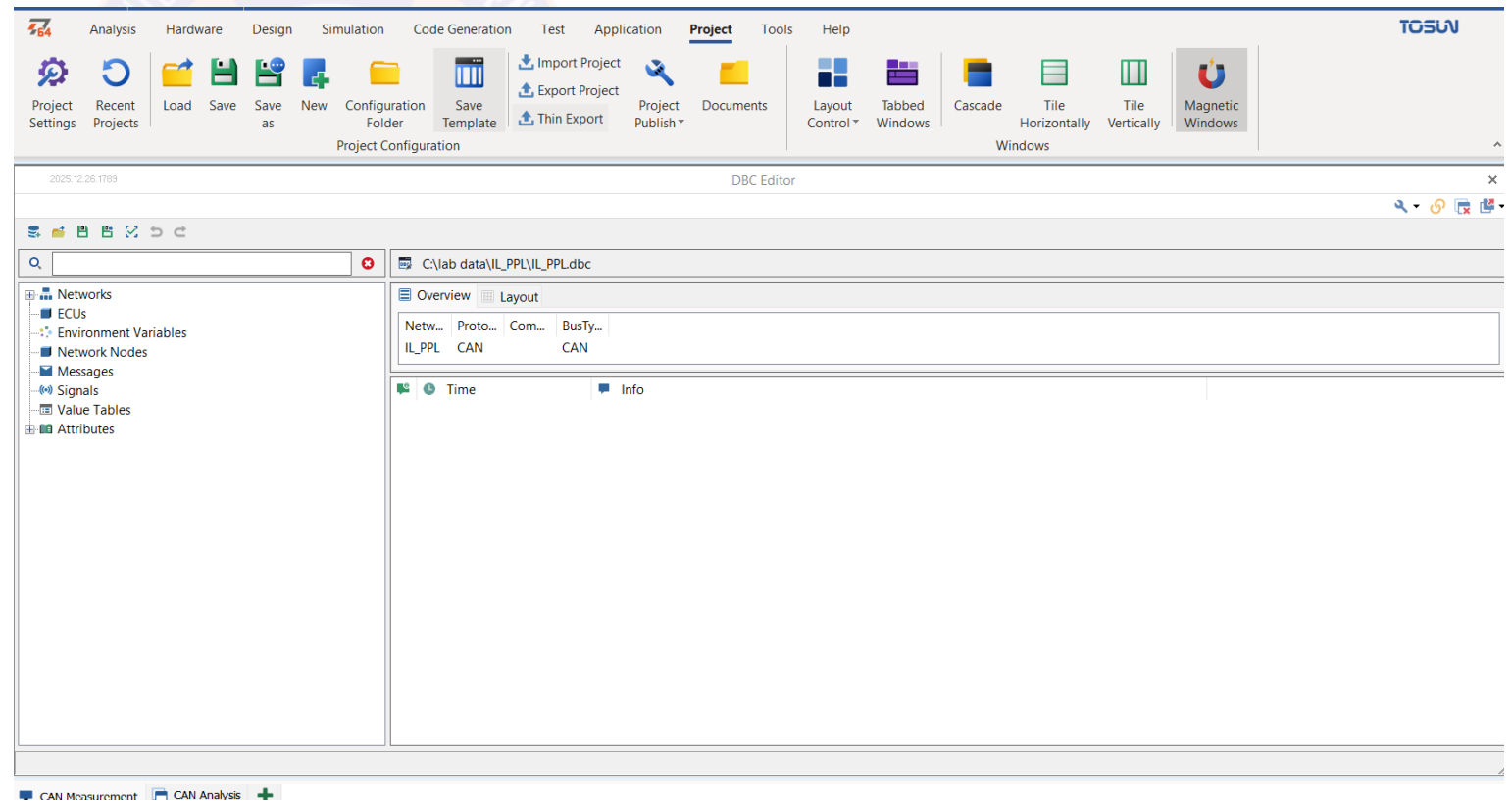
❑ Select DBC Template

❑ Click Confirm

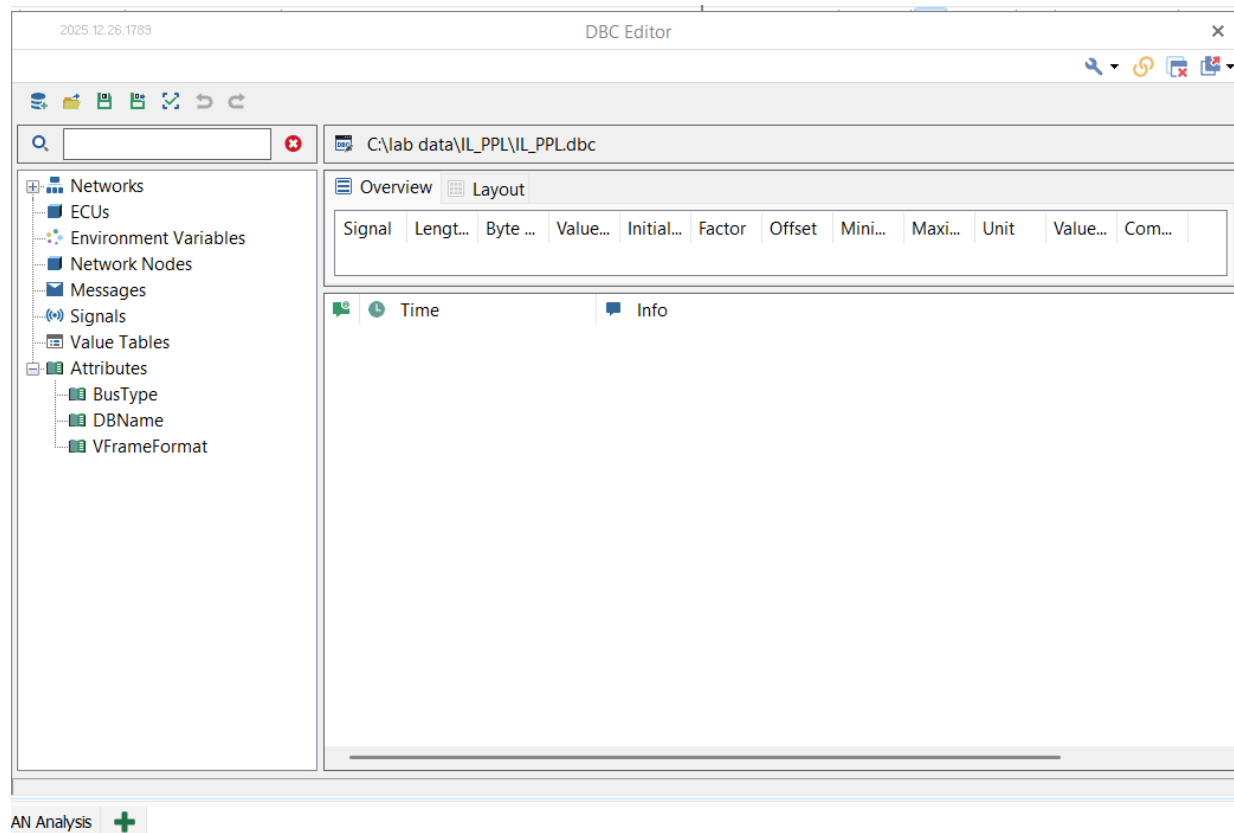


# TS MASTER

- ❑ Give a name to the DBC file and click ok
- ❑ DBC editor window will open as shown in the figure.

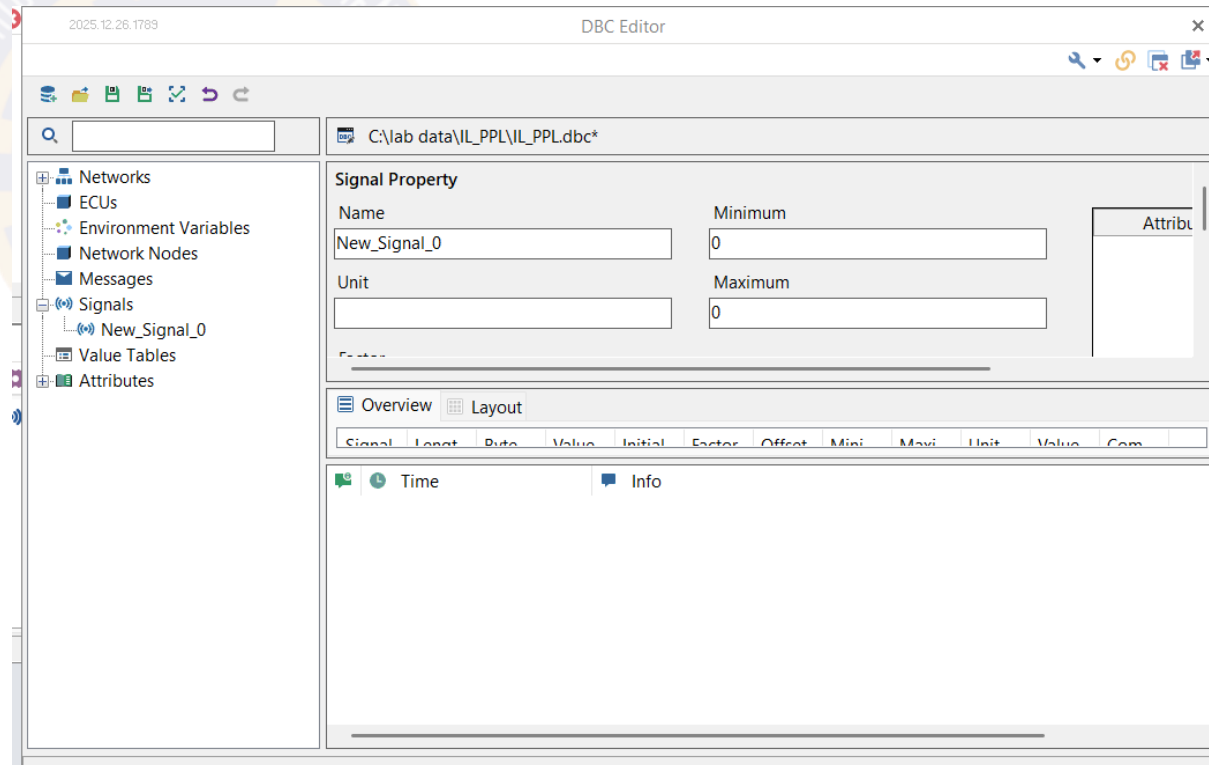


# TS MASTER



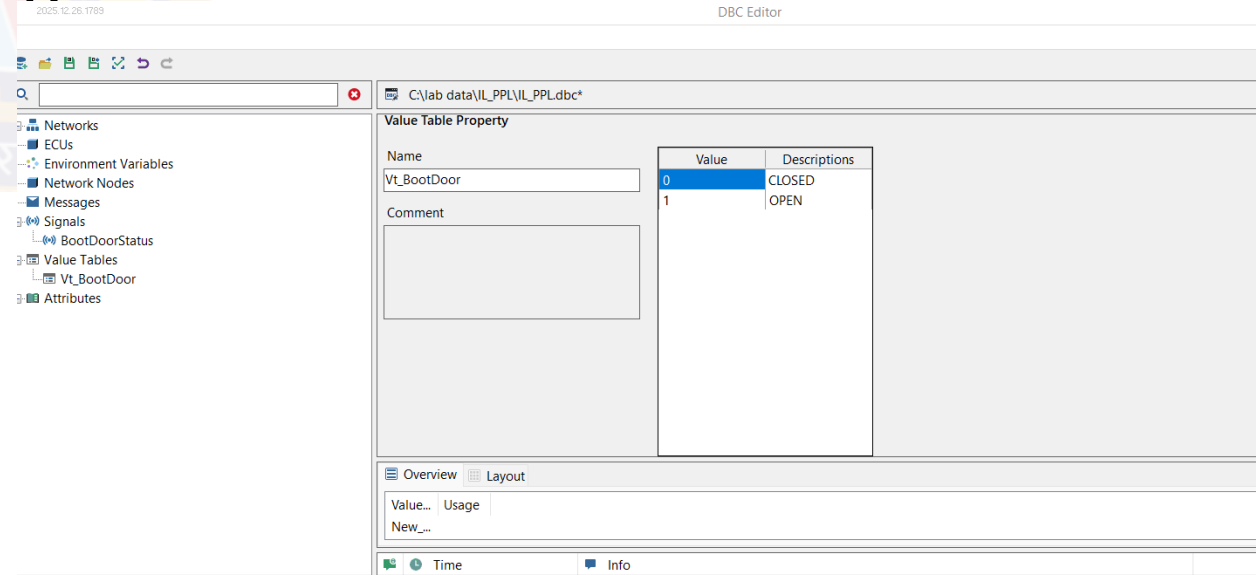
# TS MASTER

- ❑ Right click on signals and click new.
- ❑ Rename the signal as **BootDoorStatus**



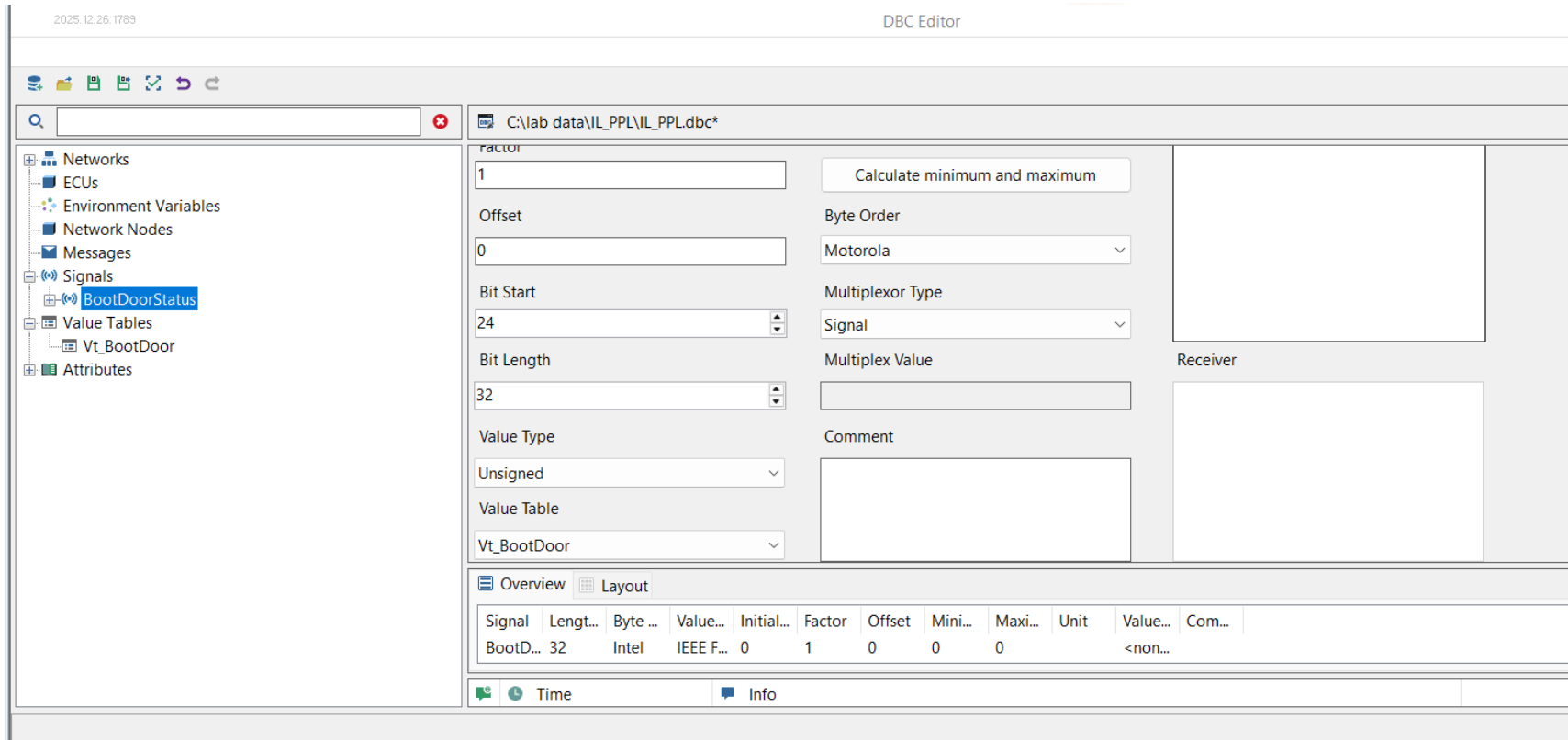
# TS MASTER

- ❑ Right click on value table and click new.
- ❑ Rename it to Vt\_BootDoor.
- ❑ Right click on the value and click new
- ❑ Type the descriptions as shown in the figure.



# TS MASTER

- ❑ Map the value table with the signal as shown in the figure.





# TS MASTER

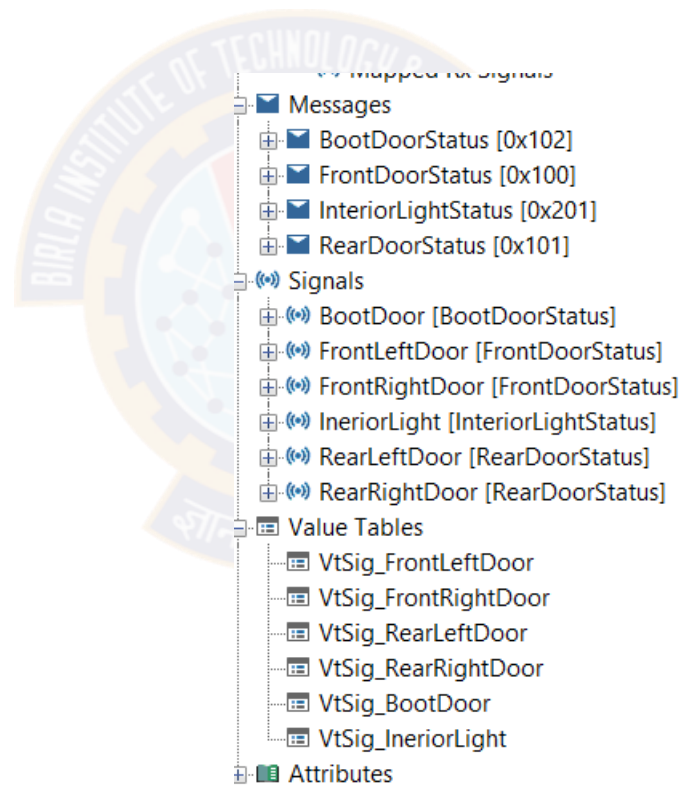
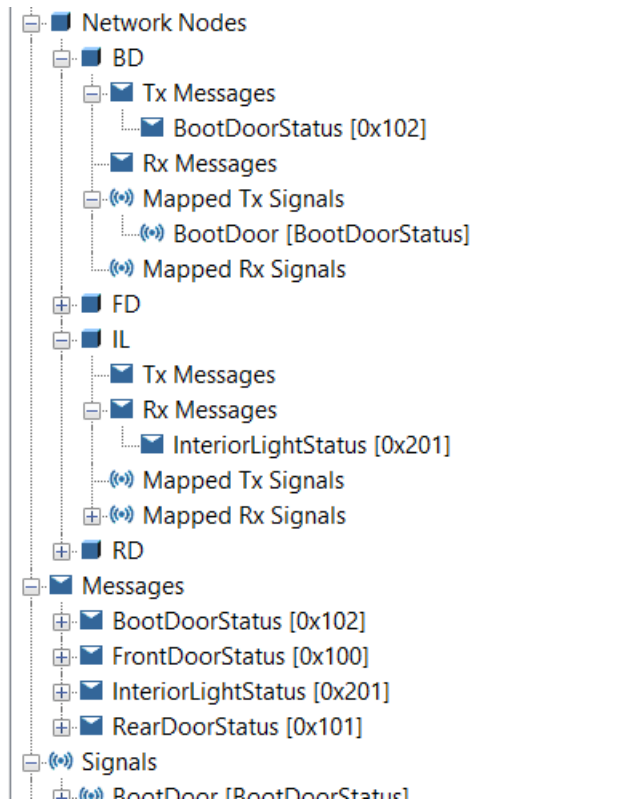
- ❑ Similarly create five signals **BootDoorStatus**, **FrontLeftDoorStatus**, **FrontRightDoorStatus**, **RearLeftDoorStatus**, **RearRightDoorStatus**, **InteriorLight**.
- ❑ Map the five signals with the respective value tables **Vt\_BootDoor**, **Vt\_FrontLeftDoor**, **Vt\_FrontRightDoor**, **Vt\_RearLeftDoor**, **Vt\_RearRightDoor**, **Vt\_InteriorLight**

# TS MASTER

- ❑ **Create a message – FrontDoor with message ID 0x100.**
  - ❑ **Drag and drop the FrontLeftDoorStatus and Front RightDoorStatus signals and map them to the FrontDoor Message**
  - ❑ **Similarly create RearDoor-0x101, BootDoor-0x102 and InteriorLight-0x201 messages and map them with the corresponding signals**
  - ❑ **Create Nodes – FD, RD,BD,IL**
  - ❑ **Now mark the messages FrontDoor, RearDoor, BootDoor as transmitter**
- And InteriorLight signal as a receiver.**

# TS MASTER

❑ The final network tree should be as shown in the image.



**Thank You**

